

#Alignment (Global) Successfully Completed
#Combined Scoring Matrix Created Using The Following
Matrix1 : BLOSUM62 Weight: 100.0%
Matrix2 : HYDROPHOBICITY Weight: 0.0%
Matrix3 : SIZE Weight: 0.0%
#Seq1 Length: 482
#Seq2 Length: 575
#GapOpening Penalty: -10
#GapExtension Penalty: -1
#Number of Alignments Found: 2
#Number of Alignments Displayed: 2

>>Rank.1 Global Alignment Score:2349.0 Sum:731564.0 Identity:99.6%

```
seq1: 1 -SGRPRT-----TCSR DGSKITTVVATPGQGTDRVQEVSYDTKVI
      |||||
seq2: 1 MSGRPRTSSFAEGNKQSPSLVLGGVKTC SR DGSKITTVVATPGQGTDRVQEVSYDTKVI

seq1: 41 GNGSFGVVFQAKLCDTGELVAIKKVLQDRRFKNRELQIMRKLEHCNIVKLLYFFYSSGEK
      |||||
seq2: 61 GNGSFGVVFQAKLCDTGELVAIKKVLQDRRFKNRELQIMRKLEHCNIVKLLYFFYSSGEK

seq1: 101 RDEVFLNLVLEYIPETVYKVAR-----FIRLYMYQLFRSLAYIHSLGICHRDI
      |||||
seq2: 121 RDEVFLNLVLEYIPETVYKVARQYAKTKQTIPINFIRLYMYQLFRSLAYIHSLGICHRDI

seq1: 149 KPQNLLLDPETAVLKLCDFGSAKQLLHGEPNVSYICSRYYRAPELIFGAINYTTKIDVWS
      |||||
seq2: 181 KPQNLLLDPETAVLKLCDFGSAKQLLHGEPNVSYICSRYYRAPELIFGAINYTTKIDVWS

seq1: 209 AGCVLAELLGQPIFPGD SGVDQLVEVIKVLGTP TREQIREMNP NYTEFKFPQIKSHPWQ
      |||||
seq2: 241 AGCVLAELLGQPIFPGD SGVDQLVEVIKVLGTP TREQIREMNP NYTEFKFPQIKSHPWQ

seq1: 269 KVFRI RTPTEAINLVSL LLEYTP SARITPLKACAHPFFDEL RMEGNHTLPNGRDM PPLFN
      |||||
seq2: 301 KVFRI RTPTEAINLVSL LLEYTP SARITPLKACAHPFFDEL RMEGNHTLPNGRDM PPLFN

seq1: 329 FTEHEL SIQPSLVPQLLPKHLQNASGPGGNRPSAGGAASIAASGSTSVSSTGSGASVEGS
      |||||
seq2: 361 FTEHEL SIQPSLVPQLLPKHLQNASGPGGNRPSAGGAASIAASGSTSVSSTGSGASVEGS

seq1: 389 AQPQS QGTAAAAGSGSGGATAGTGGASAGGPGSGN NSSSGGASGAPS AVAAGGANA AVAG
      |||||
seq2: 421 AQPQS QGTAAAAGSGSGGATAGTGGASAGGPGSGN NSSSGGASGAPS AVAAGGANA AVAG

seq1: 449 GAGGGGGAGAATAAATATGAIGATNAGGANVT-----
      |||||
seq2: 481 GAGGGGGAGAATAAATATGAIGATNAGGANVTGSQNSALNSSGSGSGNGEAAAGSGSGS

seq1: 481 -----DS
                        **
seq2: 541 GSGSGGGNGGDNDAGDSGAIASGGGAAETEAASG
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>>Rank.2 Global Alignment Score:2349.0 Sum:731213.0 Identity:99.6%

```
seq1: 1 -SGRPRT-----TCSR DGSKITTVVATPGQGTDRVQEVSYDTKVI
```

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      |||||
seq2: 1 MSGRPRTSSFAEGNKQSPSLVLGGVKTCSRDSKITTVVATPGQGTDRVQEVSYTDTKVI
      |||||
seq1: 41 GNGSFGVVFQAKLCDTGELVAIKKVLQDRRFKNRELQIMRKLEHCNIVKLLYFFYSSGEK
      |||||
seq2: 61 GNGSFGVVFQAKLCDTGELVAIKKVLQDRRFKNRELQIMRKLEHCNIVKLLYFFYSSGEK
      |||||
seq1: 101 RDEVFLNLVLEYIPETVYKVAR-----FIRLYMYQLFRSLAYIHSLGICHRDI
      |||||
seq2: 121 RDEVFLNLVLEYIPETVYKVARQYAKTKQTIPINFIRLYMYQLFRSLAYIHSLGICHRDI
      |||||
seq1: 149 KPQNLLDPETAVLKLCDFGSAKQLLHGEPNVSYICSRYYRAPELIFGAINYTTKIDVWS
      |||||
seq2: 181 KPQNLLDPETAVLKLCDFGSAKQLLHGEPNVSYICSRYYRAPELIFGAINYTTKIDVWS
      |||||
seq1: 209 AGCVLAELLGQPIFPGDSGVDQLVEVIKVLGTPTREQIREMNPNYTEFKFPQIKSHPWQ
      |||||
seq2: 241 AGCVLAELLGQPIFPGDSGVDQLVEVIKVLGTPTREQIREMNPNYTEFKFPQIKSHPWQ
      |||||
seq1: 269 KVFRIRTPTEAINLVSLLEYPSTARSITPLKACAHPPFDELMEGNHTLPNGRDMPPPLFN
      |||||
seq2: 301 KVFRIRTPTEAINLVSLLEYPSTARSITPLKACAHPPFDELMEGNHTLPNGRDMPPPLFN
      |||||
seq1: 329 FTEHELSIQPSLVPQLLPKHLQNASGPGGNRPSAGGAASIAASGSTSVSSTGSGASVEGS
      |||||
seq2: 361 FTEHELSIQPSLVPQLLPKHLQNASGPGGNRPSAGGAASIAASGSTSVSSTGSGASVEGS
      |||||
seq1: 389 AQPQSQGTAAAAGSGSGGATAGTGGASAGGPGSGNNSSSGGASGAPSAVAAGGANAAGVAG
      |||||
seq2: 421 AQPQSQGTAAAAGSGSGGATAGTGGASAGGPGSGNNSSSGGASGAPSAVAAGGANAAGVAG
      |||||
seq1: 449 GAGGGGGAGAATAAATATGAIGATNAGGANV-----
      |||||
seq2: 481 GAGGGGGAGAATAAATATGAIGATNAGGANVTGSQNSALNSSGSGSGNGEAAAGSGSGS
      |||||
seq1: 480 -----T-----DS
      |||||
seq2: 541 GSGSGGGNGGDNDAGDSGAIASGGGAAETEAASG

```

```

#Alignment (Local) Successfully Completed
#Combined Scoring Matrix Created Using The Following
# Matrix1 : BLOSUM62 Weight: 100.0%
# Matrix2 : HYDROPHOBICITY Weight: 0.0%
# Matrix3 : SIZE Weight: 0.0%
#Seq1 Length: 482
#Seq2 Length: 575
#GapOpening Penalty: -10
#GapExtension Penalty: -1
#Number of Alignments Found: 1
#Number of Alignments Displayed: 1

```

```
>>Rank.1 Local Alignment Score:2432.0 Sum:591727.0 Identity:99.8%
```

```
seq1: 1 SGRPRT-----TCSR DGSKITTVVATPGQGTDRVQEVS YTDTKVIG
      |||||
seq2: 2 SGRPRTSSFAEGNKQSPSLVLGGVKTC SRD GSKITTVVATPGQGTDRVQEVS YTDTKVIG
      |||||
seq1: 42 NGSFGVVFQAKLCDTGELVAIKKVLQDRRFKNRELQIMRKLEHCNIVKLLYFFYSSGEKR
      |||||
seq2: 62 NGSFGVVFQAKLCDTGELVAIKKVLQDRRFKNRELQIMRKLEHCNIVKLLYFFYSSGEKR
      |||||
seq1: 102 DEVFLNLVLEYIPETVYKVAR-----FIRLYMYQLFRSLAYIHSLGICH RDIK
      |||||
seq2: 122 DEVFLNLVLEYIPETVYKVARQYAKTKQTIPINFIRLYMYQLFRSLAYIHSLGICH RDIK
      |||||
seq1: 150 PQNLLLDPETAVLKLCDFGSAKQL LHGEPNVSYICSRYYRAPELIFGAINYTTKIDVWSA
      |||||
seq2: 182 PQNLLLDPETAVLKLCDFGSAKQL LHGEPNVSYICSRYYRAPELIFGAINYTTKIDVWSA
      |||||
seq1: 210 GCVLAELLLGQPIFP GDSGVDQLVEVIKVLGTP TREQIREMNP NYTEFKFPQIKSHPWQK
      |||||
seq2: 242 GCVLAELLLGQPIFP GDSGVDQLVEVIKVLGTP TREQIREMNP NYTEFKFPQIKSHPWQK
      |||||
seq1: 270 VFRIRTPTEAINLVSL LLEYTPSARITPLKACAH PFFDELRMEGNHTLPNGRDM PPLFNF
      |||||
seq2: 302 VFRIRTPTEAINLVSL LLEYTPSARITPLKACAH PFFDELRMEGNHTLPNGRDM PPLFNF
      |||||
seq1: 330 TEHEL SIQPSLVPQLLPKHLQNASGP GGNRPSAGGAASIAASGSTSVSSTGSGASVEGSA
      |||||
seq2: 362 TEHEL SIQPSLVPQLLPKHLQNASGP GGNRPSAGGAASIAASGSTSVSSTGSGASVEGSA
      |||||
seq1: 390 QPQS QGTAAAAGSGSGGATAGTGGASAGGP GSGN SSSGGASGAPS AVAAGGANAAVAGG
      |||||
seq2: 422 QPQS QGTAAAAGSGSGGATAGTGGASAGGP GSGN SSSGGASGAPS AVAAGGANAAVAGG
      |||||
seq1: 450 AGGGGGAGAATAAATATGAIGATNAGGANVTDS
      |||||*|
seq2: 482 AGGGGGAGAATAAATATGAIGATNAGGANVTGS
```